

Cloud Computing

Lab-4

Name: Monisha Sharma

SRN: PES2UG23CS906

Class: 6F

Date: 05/02/2026

SS1

```
C:\Users\Monisha>wsl
mona@MonasLaptop:~/mnt/c/Users/Monisha$ cd
mona@MonasLaptop:~$ cd PES2UG23CS906
mona@MonasLaptop:~/PES2UG23CS906$ docker --version
Docker version 29.1.3, build f52814d
mona@MonasLaptop:~/PES2UG23CS906$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

mona@MonasLaptop:~/PES2UG23CS906$ |
```

SS2

```
mona@MonasLaptop:~/PES2UG23CS906$ docker-compose --version
Docker Compose version v5.0.1
mona@MonasLaptop:~/PES2UG23CS906$
```

SS3

```

mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$ sudo docker ps
[sudo] password for mona:
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
3920a9cee253   rabbitmq:3-management              "docker-entrypoint.s..." 17 seconds ago Up 16 seconds 4369/tcp, 5671-5672/tcp
, 15671-15672/tcp, 15691-15692/tcp, 25672/tcp rabbitmq
mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$

```

SS4

```

mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$ docker run -it --name user-a chat-app

--- No Previous Chat History ---

--- Chat Started (unknown-user connecting to localhost) ---
RabbitMQ not ready, retrying... (1/5)
RabbitMQ not ready, retrying... (1/5)
RabbitMQ not ready, retrying... (2/5)
RabbitMQ not ready, retrying... (2/5)
RabbitMQ not ready, retrying... (3/5)
RabbitMQ not ready, retrying... (3/5)
RabbitMQ not ready, retrying... (4/5)
RabbitMQ not ready, retrying... (4/5)
RabbitMQ not ready, retrying... (5/5)
RabbitMQ not ready, retrying... (5/5)

ERROR: Could not connect to RabbitMQ at localhost

ERROR: Could not connect to RabbitMQ at localhost

Falling back to standalone mode...
=====
STANDALONE MODE (No RabbitMQ)
=====

--- No Previous Chat History ---

Messages will not be sent to any other container.

Type message (standalone): hellooooo hiiii
Sent by unknown-user (standalone): hellooooo hiiii
Type message (standalone): whyyy
Sent by unknown-user (standalone): whyyy
Type message (standalone): >?
Sent by unknown-user (standalone): >?
Type message (standalone): ^C
Exiting standalone mode...

```

SS5

```
mona@MonasLaptop: ~/PES x mona@MonasLaptop: ~/PES x + v
Type message (standalone): hellooooo hiii
Sent by unknown-user (standalone): hellooooo hiii
Type message (standalone): whyyy
Sent by unknown-user (standalone): whyyy
Type message (standalone): >?
Sent by unknown-user (standalone): >?
Type message (standalone): ^C
Exiting standalone mode...
mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$ docker network create chat-net
6fdea6f61978d2187550ef31751495b99d67272f41b617069dfb635f448fe0d7
mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$ docker stop user-a rabbitmq
docker rm user-a rabbitmq
user-a
rabbitmq
user-a
rabbitmq
mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$ docker run -d --name rabbitmq --network chat-net rabbitmq:3-management
9664aad2f1b0099bc53bc17f41cbd20d55f0a053cc74d7ac0aba67a329d845c9
mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$ docker run -it --name user-a \
--network chat-net \
-e RABBIT_HOST=rabbitmq \
-e QUEUE_NAME=queue_a \
-e TARGET_QUEUE=queue_b \
chat-app

--- No Previous Chat History ---

--- Chat Started (unknown-user connecting to rabbitmq) ---
Type message: hiii
Sent by unknown-user: hiii
Type message:
Received from unknown-user: hellooo
Type message: I AM USER-A
Sent by unknown-user: I AM USER-A
Type message:
Received from unknown-user: I AM USER-B
Type message:
```

SS6

```
mona@MonasLaptop: ~/PES x mona@MonasLaptop: ~/PES x + v
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Monisha>wsl
mona@MonasLaptop:/mnt/c/Users/Monisha$ cd
mona@MonasLaptop:~$ cd PES2UG23CS906/docker-advanced-lab
mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$ docker run -it --name user-b \
--network chat-net \
-e RABBIT_HOST=rabbitmq \
-e QUEUE_NAME=queue_b \
-e TARGET_QUEUE=queue_a \
chat-app

--- No Previous Chat History ---

--- Chat Started (unknown-user connecting to rabbitmq) ---
Type message:
Received from unknown-user: hiii
Type message: hellooo
Sent by unknown-user: hellooo
Type message:
Received from unknown-user: I AM USER-A
Type message: I AM USER-B
Sent by unknown-user: I AM USER-B
Type message:
```

SS7

```
mona@MonasLaptop: ~/PES x mona@MonasLaptop: ~/PES; x + v
mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$ docker stop user-a user-b rabbitmq
user-a
user-b
rabbitmq
mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$ docker rm user-a user-b rabbitmq
user-a
user-b
rabbitmq
mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$ docker network rm chat-net
chat-net
mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$
```

SS8

```
mona@MonasLaptop: ~/PES x mona@MonasLaptop: ~/PES; x + v
rabbitmq-1 | 2026-02-05 08:24:33.128830+00:00 [info] <0.254.0> Running boot step virtual_host_reconciliation defined by app rabbit
rabbitmq-1 | 2026-02-05 08:24:33.129529+00:00 [info] <0.254.0> Running boot step direct_client defined by app rabbit
rabbitmq-1 | 2026-02-05 08:24:33.129748+00:00 [info] <0.254.0> Running boot step rabbit_federation_exchange defined by app rabbitmq_
federation
rabbitmq-1 | 2026-02-05 08:24:33.133106+00:00 [info] <0.254.0> Running boot step rabbit_management_load_definitions defined by app r
abbitmq_management
rabbitmq-1 | 2026-02-05 08:24:33.134115+00:00 [info] <0.688.0> Resetting node maintenance status
rabbitmq-1 | 2026-02-05 08:24:33.715932+00:00 [warning] <0.717.0> Deprecaded features: 'management_metrics_collection': Feature 'man
agement_metrics_collection' is deprecated.
rabbitmq-1 | 2026-02-05 08:24:33.715932+00:00 [warning] <0.717.0> By default, this feature can still be used for now.
rabbitmq-1 | 2026-02-05 08:24:33.715932+00:00 [warning] <0.717.0> Its use will not be permitted by default in a future minor RabbitM
Q version and the feature will be removed from a future major RabbitMQ version; actual versions to be determined.
rabbitmq-1 | 2026-02-05 08:24:33.715932+00:00 [warning] <0.717.0> To continue using this feature when it is not permitted by default
, set the following parameter in your configuration:
rabbitmq-1 | 2026-02-05 08:24:33.715932+00:00 [warning] <0.717.0> "deprecated_features.permit.management_metrics_collection = tr
ue"
rabbitmq-1 | 2026-02-05 08:24:33.715932+00:00 [warning] <0.717.0> To test RabbitMQ as if the feature was removed, set this in your c
onfiguration:
rabbitmq-1 | 2026-02-05 08:24:33.715932+00:00 [warning] <0.717.0> "deprecated_features.permit.management_metrics_collection = fa
lse"
rabbitmq-1 | 2026-02-05 08:24:43.179245+00:00 [info] <0.754.0> Management plugin: HTTP (non-TLS) listener started on port 15672
rabbitmq-1 | 2026-02-05 08:24:43.179564+00:00 [info] <0.784.0> Statistics database started.
rabbitmq-1 | 2026-02-05 08:24:43.179671+00:00 [info] <0.783.0> Starting worker pool 'management_worker_pool' with 3 processes in it
rabbitmq-1 | 2026-02-05 08:24:43.196532+00:00 [info] <0.882.0> Prometheus metrics: HTTP (non-TLS) listener started on port 15692
rabbitmq-1 | 2026-02-05 08:24:43.197007+00:00 [info] <0.688.0> Ready to start client connection listeners
rabbitmq-1 | 2026-02-05 08:24:43.203237+00:00 [info] <0.846.0> started TCP listener on [::]:5672
rabbitmq-1 | 2026-02-05 08:24:43.625966+00:00 [info] <0.688.0> Server startup complete; 5 plugins started.
rabbitmq-1 | 2026-02-05 08:24:43.625966+00:00 [info] <0.688.0> * rabbitmq_prometheus
rabbitmq-1 | 2026-02-05 08:24:43.625966+00:00 [info] <0.688.0> * rabbitmq_federation
rabbitmq-1 | 2026-02-05 08:24:43.625966+00:00 [info] <0.688.0> * rabbitmq_management
rabbitmq-1 | 2026-02-05 08:24:43.625966+00:00 [info] <0.688.0> * rabbitmq_management_agent
rabbitmq-1 | 2026-02-05 08:24:43.625966+00:00 [info] <0.688.0> * rabbitmq_web_dispatch
rabbitmq-1 | completed with 5 plugins.
rabbitmq-1 | 2026-02-05 08:24:43.798108+00:00 [info] <0.9.0> Time to start RabbitMQ: 25720 ms

View in Docker Desktop View Config Enable Watch Detach
```

SS9

```

mona@MonasLaptop: ~/PES: X      mona@MonasLaptop: ~/PES: X      mona@MonasLaptop: ~/PES: X      mona@MonasLaptop: ~/PES: X
mona@MonasLaptop: /mnt/c/Users/Monisha$ cd ~/PES2UG23CS906/docker-advanced-lab
mona@MonasLaptop: ~/PES2UG23CS906/docker-advanced-lab$ docker compose run --rm user-a
WARN[0000] /home/mona/PES2UG23CS906/docker-advanced-lab/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential c
onfusion
WARN[0000] No services to build
[*] 1/1t 1/1t
✓Container docker-advanced-lab-rabbitmq-1 Running      0.0s
Image docker-advanced-lab-user-a Building
[*] Building 4/4s (13/13) FINISHED
  => [internal] load local base definitions      0.0s
  => reading from stdin 564B      0.0s
  => [internal] load build definition from Dockerfile      0.1s
  => transferring dockerfile: 32B      0.0s
  => [internal] load metadata for docker.io/library/python:3.9-slim      2.9s
  => [auth] library/python:pull token for registry-1.docker.io      0.0s
  => [internal] load dockerignore      0.0s
  => transferring context: 2B      0.0s
  => [1/1] FROM docker.io/library/python:3.9-slimsha256:2d976910b16db3db3068f261f53f1040665785599aaabacd1e13c1f731b1b      0.1s
  => resolve docker.io/library/python:3.9-slimsha256:2d976910b16db3db3068f261f53f1040665785599aaabacd1e13c1f731b1b      0.1s
  => [internal] load build context      0.0s
  => transferring context: 20B      0.0s
  => CACHED [2/5] RUN pip install piwa      0.0s
  => CACHED [3/5] WORKDIR /app      0.0s
  => CACHED [4/5] COPY chat.py      0.0s
  => CACHED [5/5] RUN mkdir -p /app/data      0.0s
  => exporting to image      0.4s
  => exporting layers      0.0s
  => exporting manifest sha256:aa80890d2f481f4440d72dfbb7dcb8d84ccdc5db8d444d3f367c3edaa6f3f93      0.0s
  => exporting config sha256:756b2f08fb055e18856569af1db137d4857fc8a6313ac5f4240d9075b9527ac      0.1s
  => exporting attestation manifest sha256:520aa90b2027577ad5f66630dc63c9424f0466f37033f7bb27cac222aaef14      0.1s
  => exporting manifest list sha256:a09c9e953c8a3b1bd9fb2527f9060f604bf9a81171ad6e071ec9771b19385      0.0s
  => naming to docker.io/library/docker-advanced-lab-user-a:latest      0.0s
  => unpacking to docker.io/library/docker-advanced-lab-user-a:latest      0.0s
  => resolving provenance for metadata file      0.0s
Image docker-advanced-lab-user-a Built
Container docker-advanced-lab-user-a-run-e02b1d0e9a42 Creating
Container docker-advanced-lab-user-a-run-e02b1d0e9a42 Created

--- No Previous Chat History ---

--- Chat Started (user-a connecting to rabbitmq) ---
Type message: HELLO
Sent by user-a: HELLO
Type message:
Received from user-b: HI!!!
Type message: I AM USER-A
Sent by user-a: I AM USER-A
Type message:
Received from user-b: I AM USER-B
Type message:

```

SS10

```

mona@monasLaptop: ~/PES : x      mona@monasLaptop: ~/PES : x      mona@monasLaptop: ~/PES : x      mona@monasLaptop: ~/PES : x
rary/python/manifests/3.9-slim": tls: failed to verify certificate: x509: certificate signed by unknown authority

mona@monasLaptop:~/PES2UG23C5906/docker-advanced-lab$ docker compose run --rm user-b
WARN[0000] /home/mona/PES2UG23C5906/docker-advanced-lab/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential c
onfusion
WARN[0000] No services to build
[*] 1/it 1/11
✓Container docker-advanced-lab-rabbitmq-1 Running 0.0s
Image docker-advanced-lab-user-b Building 0.0s
[*] Building 2.2s (12/12) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 564B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 323s 0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 1.2s
=> [internal] load dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/1] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd33d38d3806f261f43f144965f75599aabb4ed413cf1731b1b 0.2s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd33d38d3806f261f43f144965f75599aabb4ed413cf1731b1b 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 29B 0.0s
=> CACHED [2/5] RUN pip install pika 0.0s
=> CACHED [3/5] WORKDIR /app 0.0s
=> CACHED [4/5] COPY chat.py 0.0s
=> CACHED [5/5] RUN mkdir -p /app/data 0.0s
=> exporting to image 0.3s
=> => exporting layers 0.0s
=> => exporting manifest sha256:9809f3bb1169a35106286fbdcc1c73105ea0e2a0c947272c67ba2fa8c6cf518c 0.0s
=> => exporting config sha256:531464a6796609898239be110103a05fee11bf1f248d3e15dbdc6c6c846be563f 0.0s
=> => exporting attestation manifest sha256:d42d2d2f73720f5d62a06d28abbf5f54e8b10d3c1fc72840e92472f9d3d7acc 0.0s
=> => exporting manifest list sha256:948ddaf28b314d4d1364bd20c298d99804d6a13f0bc225a3b917b716afffbaad 0.0s
=> naming to docker.io/library/docker-advanced-lab-user-b:latest 0.0s
=> => unpacking to docker.io/library/docker-advanced-lab-user-b:latest 0.0s
=> resolving provenance for metadata file 0.0s
Image docker-advanced-lab-user-b Built
Container docker-advanced-lab-user-b-run-620eb35152b3 Creating
Container docker-advanced-lab-user-b-run-620eb35152b3 Created

--- No Previous Chat History ---

--- Chat Started (user-b connecting to rabbitmq) ---
Type message:
Received from user-a: HELLO
Type message: HIIII
Sent by user-b: HIIII
Type message:
Received from user-a: I AM USER-A
Type message: I AM USER-B
Sent by user-b: I AM USER-B
Type message:

```

SS11

```
mona@MonasLaptop: ~/PES: x  mona@MonasLaptop: ~/PES: x  mona@MonasLaptop: ~/PES: x  + v
Sent by user-a: BYE
Type message:
Received from user-b: AU REVOIR
Type message: ^C
Chat ended.
mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$ docker compose run --rm user-a
WARN[0000] /home/mona/PES2UG23CS906/docker-advanced-lab/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored,
please remove it to avoid potential confusion
WARN[0000] No services to build
[+] 1/1t 1/11
✔Container docker-advanced-lab-rabbitmq-1 Running
0.0s
WARN[0000] No services to build
Container docker-advanced-lab-user-a-run-8a118a701608 Creating
Container docker-advanced-lab-user-a-run-8a118a701608 Created

--- Chat History ---
Sent by user-a: AAA
Received from user-a: AAA
Sent by user-b: BBB
Received from user-b: BBB
Sent by user-a: CCC
Received from user-a: CCC
Sent by user-b: DDD
Received from user-b: DDD
Sent by user-a: BYE
Received from user-a: BYE
Sent by user-b: AU REVOIR
Received from user-b: AU REVOIR
--- End of History ---

--- Chat Started (user-a connecting to rabbitmq) ---
Type message: NICE TO MEET U
Sent by user-a: NICE TO MEET U
Type message:
Received from user-b: SAME HERE
Type message:
```

SS12

```
mona@MonasLaptop: ~/PES: x  mona@MonasLaptop: ~/PES: x  mona@MonasLaptop: ~/PES: x  + v
Sent by user-b: AU REVOIR
Type message: ^C
Chat ended.
mona@MonasLaptop:~/PES2UG23CS906/docker-advanced-lab$ docker compose run --rm user-b
WARN[0000] /home/mona/PES2UG23CS906/docker-advanced-lab/docker-compose.yml: the attribute 'version' is obsolete, it will be
ignored, please remove it to avoid potential confusion
WARN[0000] No services to build
[+] 1/1t 1/11
✔Container docker-advanced-lab-rabbitmq-1 Running
0.0s
WARN[0000] No services to build
Container docker-advanced-lab-user-b-run-f34bf3bd8aa6 Creating
Container docker-advanced-lab-user-b-run-f34bf3bd8aa6 Created

--- Chat History ---
Sent by user-a: AAA
Received from user-a: AAA
Sent by user-b: BBB
Received from user-b: BBB
Sent by user-a: CCC
Received from user-a: CCC
Sent by user-b: DDD
Received from user-b: DDD
Sent by user-a: BYE
Received from user-a: BYE
Sent by user-b: AU REVOIR
Received from user-b: AU REVOIR
--- End of History ---

--- Chat Started (user-b connecting to rabbitmq) ---
Type message:
Received from user-a: NICE TO MEET U
Type message: SAME HERE
Sent by user-b: SAME HERE
Type message: |
```